



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

COS 226 - Concurrent Systems

Practical 1 Specification

Mutual Exclusion Algorithms

Release Date: Monday, 27 July 2026

Submission Deadline: Sunday, 2 August 2026

1 Overview

The purpose of this practical is to develop an understanding of classical software mutual exclusion algorithms.

You are expected to implement three locking algorithms, demonstrate their behaviour using two concurrent threads, and explain the concepts behind each algorithm during your practical demonstration.

The emphasis of this practical is on understanding the algorithms rather than simply producing working code.

2 Learning Outcomes

By completing this practical you should be able to:

- Explain the purpose of mutual exclusion.
- Implement software locking algorithms.
- Demonstrate the behaviour of concurrent threads.
- Explain why certain algorithms fail.
- Explain how Peterson's Lock resolves these failures.

3 Requirements

You are free to design the structure and architecture of your solution. No starter template will be provided. Your implementation should demonstrate good software engineering practices.

This practical is to be completed in groups of **three**. Group members are expected to contribute equally to the implementation and should all be able to explain any part of the solution during the demonstration.

Your implementation must contain the following algorithms:

1. LockOne
2. LockTwo
3. Peterson's Lock

Your implementation must demonstrate the algorithms using exactly two threads.

4 Submission

Your solution will be submitted via **Clickup**. Your submission should include:

- Source code.
- A short README explaining how to run your program.
- Clear comments indicating where each required feature has been implemented.

No late submissions will be accepted!

5 Demonstration

Students are required to demonstrate their implementation to a tutor during their scheduled practical session.

During the demonstration, students must:

- Demonstrate the implementation of LockOne, LockTwo and Peterson's Lock.
- Demonstrate that their implementation operates correctly using two concurrent threads.
- Clearly identify where each required algorithm has been implemented.
- Be prepared to answer conceptual questions that assess their understanding of the algorithms and the concepts covered in this practical.

6 Marking Rubric

Assessment Item	Marks
LockOne implementation	1
LockTwo implementation	1
Peterson's Lock implementation	2
Conceptual Question 1	3
Conceptual Question 2	3
Total	10